



No. of Functional Features Included in the Solution

Date	
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Crop Recommendation Engine	Accepts soil (N, P, K, pH) and environmental (temperature, humidity, rainfall) inputs and predicts the most suitable crop using a trained Random Forest classifier (99.5% accuracy). Displays predicted crop, confidence score, and detailed crop info (ideal temperature, rainfall, pH, agricultural tips).	Frontend (recommend.html) Backend (/api/predict) ML Model (model/predict.py)	Done	

2	Suitability Analysis	Evaluates how suitable a given set of conditions is for crop cultivation. Returns suitability status (Optimal/Moderate/Poor), productivity level, environmental compatibility badges, soil analysis, rainfall/temperature analysis, and tailored recommendations.	Frontend (suitability.html) Backend (/api/suitability)	Done	
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3	Research Dashboard	Displays comprehensive dataset analytics: summary cards, model performance comparison table with bestmodel highlighting, Chart.js bar chart (accuracy/precision/recall/F1 across 6 models), dataset statistics, environmental summary, top feature correlations, K-Means analysis card, and 10 chart images.	Frontend (dashboard.html) Backend (/api/dashboard, /api/metrics, /api/images)	Done	
4	Multi-Model Training & Comparison	Trains and evaluates 6 ML classifiers (Random Forest, SVM, Decision Tree, Logistic Regression, KNN, Naive Bayes) with crossvalidation. Identifies bestperforming model. Saves metrics (accuracy, precision, recall, F1, CV mean, CV std) to JSON.	Backend (model/train_model.py)	Done	
5	K-Means Exploratory Clustering	Performs unsupervised KMeans clustering on PCAreduced dataset (2D). Visualizes clusters alongside actual crop labels. Explicitly labeled as exploratory only — not used for prediction.	Backend (model/train_model.py, utils/visualization.py)	Done	

6	Chart Image Generation	Generates 10 publicationquality chart images (crop distribution, NPK, temperature, humidity, rainfall, pH, correlation heatmap, boxplots, pairplot, K-Means clusters). Served via API with local fallback copies in frontend.	Backend (utils/visualization.py, /api/images)	Done	
7	Static Frontend Site (6 Pages)	Multi-page static site with 6 HTML pages (Home, Recommend, Suitability, Dashboard, About, Contact). Bootstrap 5 responsive navbar, footer, CSS, form validation, error alerts, loading spinners, smooth	Frontend (6 HTML files, css/style.css, js/script.js)	Done	
		scroll, auto-dismiss alerts. Vercel-deployable.			
8	RESTful JSON API (8 Endpoints)	8 JSON endpoints: health check, crop prediction, suitability, dashboard stats, model metrics, crop info, crop list, chart images. Flask-CORS enabled. Proper HTTP status codes (200, 400, 404, 500, 503).	Backend (app.py, config.py)	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Crop Recommendation, Suitability Analysis, Dashboard, Multi-Model Training, Chart Generation, REST API)
Additional / Nice-to-Have Features	2 (K-Means Clustering, Static Frontend SPA)

Features Tested & Verified	8 (All endpoints tested via Flask test client; diverse crop predictions verified)
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Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	6	HTML pages (Home, Recommend, Suitability, Dashboard, About, Contact), responsive navbar, footer, loading spinners, alerts, Chart.js bar chart, chart image cards with fallback
2	Backend / Logic	6	Crop prediction engine, suitability assessment, multimodel training, K-Means clustering, chart generation, dashboard stats computation
3	Database / Storage	2	Dataset (Crop_recommendation.csv), model artifacts (crop_model.pkl, label_encoder.pkl, pca_data.pkl, metrics.json)
4	API / Integration	8	/api/predict, /api/suitability, /api/dashboard, /api/health, /api/metrics, /api/crop-info, /api/images, /api/crops
5	Security / Authentication	0	Not applicable for this academic/research project